Seeing Clearly

A dataviz study program, completed

Eight sources. Forty-eight rules. One standard for every figure we ship.

Research Analyst · AIMLDS (2351485 ALBERTA INC.)

Completed 3 November 2026

Why a research product studies drawing

- **The problem.** A research product's charts are claims. A truncated axis, a smoothed gap, an unlabelled band — each one is a small falsehood told at scale.
- **The directive.** On 26 September 2026 the Director set a standing study track: learn how the best in the world draw data, distill it into rules, and hold our own figures to them.
- **The honesty standard, from day one.** Missing data stays missing. Uncertainty is shown, not smoothed. Scales never lie by default. Every chart answers: what period, what source, what limitation.
- **Where it feeds.** The Quant Lab, the Research Analyst lenses, and the future Conduit dataviz product.

Our World in Data — provenance is structure

- **What it is.** The OWID Grapher: open-source charting behind Our World in Data's published research.
- **Core philosophy.** The chart is never separated from its provenance — every chart ships with its data table, its sources, and its download.
- **Most portable technique.** The chart footer contract: a title that states the finding and its period, a source line, a processing line, and a limitation line — on every chart, in that order.
- **Distilled into.** PLAYBOOK rules 1–7.

Flourish — templates carry the editorial craft

- **What it is.** A no-code chart and story builder whose templates encode editorial judgment, so the author doesn't rebuild the craft per chart.
- **Core philosophy.** Defaults carry the craft: scrollytelling and guided interactivity are templates with a published “when to use” gate.
- **Most portable technique.** Annotations state the takeaway, not the description — every important chart is paired with a caption that completes “This matters because…”.
- **Distilled into.** PLAYBOOK rules 8–12.

The Data Visualisation Catalogue — choose by question

- **What it is.** Severino Ribecca's taxonomy of ~60 chart types, each classified by the functions it serves.
- **Core philosophy.** Every chart is classified by the question it answers — and every description states what the chart cannot show.
- **Most portable technique.** The “what it hides” clause: each chart type ships with its blindness stated in the limitation line — a span chart hides everything between the extremes; a pie hides precise comparisons.
- **Distilled into.** PLAYBOOK rules 13–18.

Neo4j — a graph scene is a query result

- **What it is.** Neo4j Bloom and Browser: graph visualization over node-and-relationship data.
- **Core philosophy.** A graph picture is a query result, not the data itself — a Perspective decides what an audience sees, and search builds a small scene instead of dumping the database.
- **Most portable technique.** The data funnel runs before any layout: filter at query time, aggregate with counts, fit the visual model to the question — layout last, at roughly 100 visible nodes.
- **Distilled into.** PLAYBOOK rules 19–24.

SVG — the chart is DOM, not pixels

- **What it is.** Hand-built vector graphics: the chart as inspectable markup, no chart library in between.
- **Core philosophy.** Every mark is an element and every label is real text — so the author, not a library, owns every encoding decision, including the dishonest ones a library would have prevented.
- **Most portable technique.** The accessible-SVG pattern as the template, not a retrofit: role="img", a <title> carrying finding + period, a <desc> carrying trend, extremes, and limitation.
- **Distilled into.** PLAYBOOK rules 25–30.

Plotly — the figure is JSON, the view is state

- **What it is.** Interactive statistical charting: a figure is a declarative JSON object — traces, layout, config.
- **Core philosophy.** Interactivity is view state on top of a spec that can be saved, diffed, and re-rendered — so the author owns the default view, and what every reader-changed view will silently claim.
- **Most portable technique.** The figure spec is the artefact: store it, diff it, and lint its honesty fields — connectgaps, rangebreaks, axis range and type — before anything renders.
- **Distilled into.** PLAYBOOK rules 31–36.

Graphviz — the diagram is text, the engine is a claim

- **What it is.** Diagram layout algorithms: one plain-text DOT specification, a family of named layout engines.
- **Core philosophy.** Each engine encodes a different claim about what position means — in dot, position is rank in a flow; in twopi, radius is hops from a declared root. Swap the engine and the same file makes a different claim.
- **Most portable technique.** Name the engine in the diagram footer — position means only what the engine claims — and lint the DOT spec before rendering.
- **Distilled into.** PLAYBOOK rules 37–42.

Databricks — the dashboard is an artefact

- **What it is.** AI/BI (Lakeview) dashboards at data scale: a dashboard is a single serialized JSON artefact (.lvdash.json).
- **Core philosophy.** The dataset is the unit: datasets are named queries with their own identity in the spec, and every widget is a view of one — two panels over the same dataset cannot silently diverge.
- **Most portable technique.** Datasets as named artefacts, charts as views — and the counter contract: no big number ships unless it can name its comparison and its period.
- **Distilled into.** PLAYBOOK rules 43–48.

The PLAYBOOK — 48 rules

- **Honesty of provenance (rules 1–7, 32, 43–45).** Every chart carries title, source, processing, and limitation; a filed number and a computed number are never visually indistinguishable.
- **Honesty of scale (rules 3, 18, 26, 33).** Bars start at zero; transforms are announced in the title; scaling never distorts; axis breaks are named transforms, never silent.
- **Honesty of uncertainty (rules 2, 14, 15, 34).** Substitutions are noticed on the chart; every band names its measure; uncertainty is drawn with native geometry, never invented.
- **Honesty of position (rules 21, 37–42).** Layout is a named choice; the engine goes in the footer; proximity is never presented as evidence.
- **The spec is the artefact (rules 30, 36, 42, 48).** Figures and diagrams are text — stored, diffed, and linted before they render.

Complement, not compete

- **The principle.** Where Plotly's product is genuinely powerful, we use it — even where Plotly could be cast as a competitor. Using it signals readiness to complement in a healthy way, not compete.
- **Adopt.** Waterfall, heatmap, indicator, and box/violin traces, and the bar/scatter primitives with unified hover — as shipped, never rebuilt.
- **Adapt.** Plotly's template system carries the Ink & Brass house theme: one template, every chart, no per-chart styling code.
- **Avoid.** Custom chart types of our own — that would be competing. And Plotly's cloud services: the library is vendored and self-hosted, so the product fully works if every external service vanished tonight.
- **What stays ours.** The first-source statement data, the computed quant engine, and the provenance/honesty layer. Plotly draws; the engine decides what's true.

Where it lives now — the Quant Lab

- **Phase 1 (spec, 2 Nov 2026).** An interactive Plotly chart layer inside the analyst app's existing Quant tab, rendered from the live quant payload — trends, DuPont, forensics, valuation, allocation, expected value.
- **Additive, never replacing.** Every table the tab renders today stays, in the same order. If charting fails, the tab degrades to exactly today's tables — and the tables remain the accessible artefact.
- **The rules became acceptance criteria.** PLAYBOOK rules 31–36 are written as testable checks: a named default view, gaps that actually break the line, no invented uncertainty, missing inputs rendered visibly — never zero-filled.
- **Proof before deploy.** A Gate 17 real-browser walk of the primary flow is required before any deploy decision. File presence is not proof.

The standard for every figure

- **Source, scale, limits — stated.** No figure ships without its footer: what period, what source, what limitation.
- **Missing stays missing.** No zero-fill, no forward-fill, no silent bridging of a gap. A bridged segment is dashed and noticed, or it is not drawn.
- **Uncertainty is named.** Every band, bar, and interval states which measure it draws — or it doesn't ship.
- **The default view is declared.** A reader's zoom is an edit of the chart's claim; the chart says so, and exports render the declared view.
- **Specs are linted before rendering.** The honesty fields are machine-readable — a checker verifies them before a reader ever sees the figure.

COMPLETION

Studies complete — 8 of 8

- ✓ Our World in Data — provenance is structure
- ✓ Flourish — templates carry the craft
- ✓ The Data Visualisation Catalogue — choose by question
- ✓ Neo4j — a graph scene is a query result
- ✓ SVG — the chart is DOM, not pixels
- ✓ Plotly — the figure is JSON
- ✓ Graphviz — the engine is a claim
- ✓ Databricks — the dataset is the unit

48 rules distilled into the PLAYBOOK — the living standard for Research Analyst, the Quant Lab, and every figure that follows.

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